

NIDS Session Report

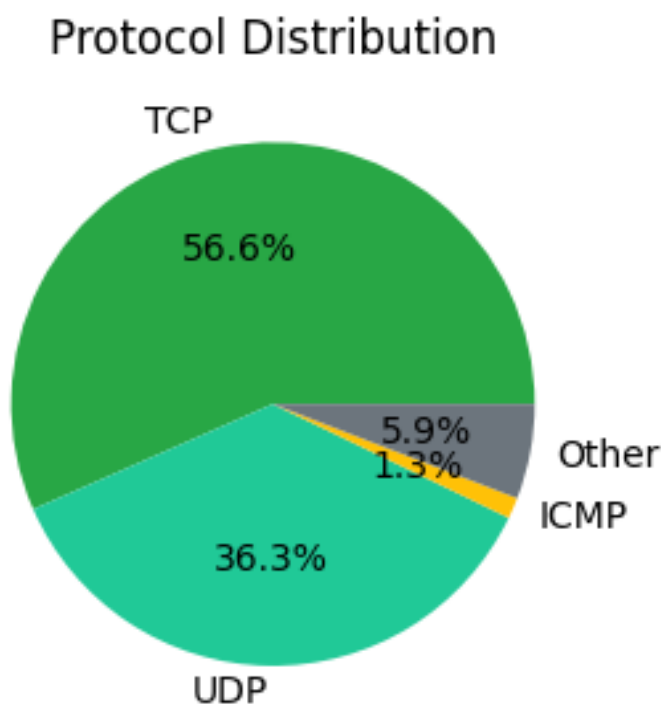
AI-Based Network Intrusion Detection System

This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Protocol Distribution



Session Summary

Start Time:	1750245192.9161036
Stop Time:	2025-06-18T16:44:32.410718
Uptime:	79 seconds
Packets Processed:	615
Anomalies Detected:	15
Packet Rate:	0 packets/sec

Protocol Statistics

TCP PACKETS:	348
UDP PACKETS:	223
ICMP PACKETS:	8
OTHER PACKETS:	36

Security Alerts (5)

Type: Port Scan | Time: 2025-06-18T16:43:14.739211

Details: {'src_ip': '10.0.2.15', 'dst_ip': '10.0.2.3', 'dst_port': 53, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: Ddos Indicator | Time: 2025-06-18T16:43:17.061212

Details: {'src_ip': '3.86.21.167', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

Type: MI Anomaly | Time: 2025-06-18T16:43:17.960580

Details: {'anomaly_score': -7.302000386468828, 'features': {'packet_count': 86, 'total_bytes': 13631, 'avg_packet_length': np.float64(158.5), 'std_packet_length': np.float64(311.1178551087061), 'min_packet_length': 54, 'max_packet_length': 1536, 'avg_payload_size': np.float64(103.1046511627907), 'std_payload_size': np.float64(312.0005973857492), 'total_payload_size': 8867, 'tcp_count': 76, 'udp_count': 10, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.8837209302325582, 'udp_ratio': 0.11627906976744186, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 16, 'unique_dst_ports': 44, 'total_ports_accessed': 44, 'common_port_ratio': 0.313953488372093, 'high_port_ratio': 0.686046511627907, 'avg_dst_port': np.float64(36510.17441860465), 'std_dst_port': np.float64(24471.672639914905), 'min_dst_port': 53, 'max_dst_port': 60137, 'unique_connections': 25, 'total_connection_attempts': 86, 'avg_packets_per_connection': 3.44, 'connection_diversity': 0.29069767441860467, 'max_packets_per_connection': 16, 'unique_src_ips': 16, 'unique_dst_ips': 11, 'src_ip_diversity': 0.18604651162790697, 'dst_ip_diversity': 0.12790697674418605, 'private_src_ip_ratio': 0.125, 'private_dst_ip_ratio': 0.18181818181818182, 'packet_rate': 19.5870046078395, 'inter_arrival_time_mean': np.float64(0.05165489701663747), 'inter_arrival_time_std': np.float64(0.11014286096378792), 'burst_count': 54, 'burst_ratio': 0.6352941176470588, 'payload_size_mean': np.float64(103.1046511627907), 'payload_size_std': np.float64(312.0005973857492), 'zero_payload_ratio': 0.1744186046511628, 'large_payload_ratio': 0.046511627906976744, 'payload_size_entropy': np.float64(0.790553396661191), 'syn_count': 44, 'fin_count': 2, 'rst_count': 0, 'psh_count': 12, 'ack_count': 68, 'urg_count': 0, 'syn_ratio': 0.5789473684210527, 'fin_ratio': 0.02631578947368421, 'rst_ratio': 0.0, 'psh_ratio': 0.15789473684210525, 'ack_ratio': 0.8947368421052632, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:43:17.960566', 'packet_count': 86, 'protocol_distribution': {'tcp': 76, 'udp': 10, 'icmp': 0}, 'connection_info': {'unique_connections': 25, 'unique_src_ips': 16, 'unique_dst_ips': 11}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Port Scan | Time: 2025-06-18T16:44:14.749088

Details: {'src_ip': '10.0.2.15', 'dst_ip': '107.20.191.65', 'dst_port': 443, 'protocol': 6}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: MI Anomaly | Time: 2025-06-18T16:44:18.118737

Details: {'anomaly_score': -7.296225567601673, 'features': {'packet_count': 100, 'total_bytes': 36664, 'avg_packet_length': np.float64(366.64), 'std_packet_length': np.float64(496.62332848951024), 'min_packet_length': 54, 'max_packet_length': 1536, 'avg_payload_size': np.float64(246.05), 'std_payload_size': np.float64(444.80648320365117), 'total_payload_size': 24605, 'tcp_count': 37, 'udp_count': 48, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.43529411764705883, 'udp_ratio': 0.5647058823529412, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 6, 'unique_dst_ports': 7, 'total_ports_accessed': 7, 'common_port_ratio': 0.4235294117647059, 'high_port_ratio': 0.5764705882352941, 'avg_dst_port': np.float64(31674.129411764705), 'std_dst_port': np.float64(26930.614433823317), 'min_dst_port': 443, 'max_dst_port': 61784, 'unique_connections': 11, 'total_connection_attempts': 85, 'avg_packets_per_connection': 7.7272727272727275, 'connection_diversity': 0.11, 'max_packets_per_connection': 16, 'unique_src_ips': 7, 'unique_dst_ips': 6, 'src_ip_diversity': 0.07, 'dst_ip_diversity': 0.06, 'private_src_ip_ratio': 0.14285714285714285, 'private_dst_ip_ratio': 0.16666666666666666, 'packet_rate': 70.37178406723122, 'inter_arrival_time_mean': np.float64(0.014353778627183702), 'inter_arrival_time_std': np.float64(0.04860445151676539), 'burst_count': 68, 'burst_ratio': 0.6868686868686869, 'payload_size_mean': np.float64(246.05), 'payload_size_std': np.float64(444.80648320365117), 'zero_payload_ratio': 0.23, 'large_payload_ratio': 0.13, 'payload_size_entropy': np.float64(1.4761457671584457), 'syn_count': 5, 'fin_count': 2, 'rst_count': 0, 'psh_count': 13, 'ack_count': 35, 'urg_count': 0, 'syn_ratio': 0.13513513513513514, 'fin_ratio': 0.05405405405405406, 'rst_ratio': 0.0, 'psh_ratio': 0.35135135135135137, 'ack_ratio': 0.9459459459459459, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:44:18.118677', 'packet_count': 100, 'protocol_distribution': {'tcp': 37, 'udp': 48, 'icmp': 0}, 'connection_info': {'unique_connections': 11, 'unique_src_ips': 7, 'unique_dst_ips': 6}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed: Total number of network packets analyzed by the NIDS during this session.

Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.